

Listening to the ground beneath IIT Guwahati

Can an ordinary phone tell which parts of our campus would shake hardest in an earthquake? We recorded five spots in one evening to find out, and learned exactly what it takes to answer the question properly.

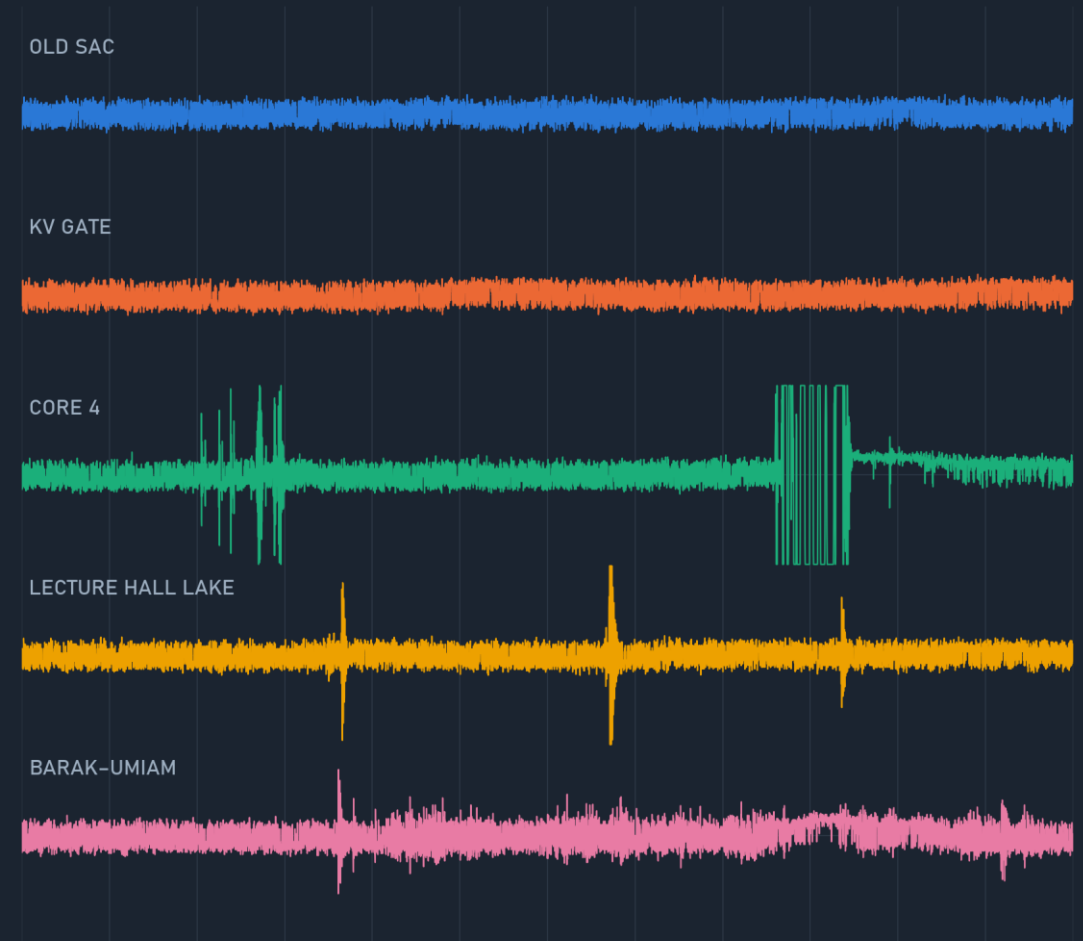
TEAM



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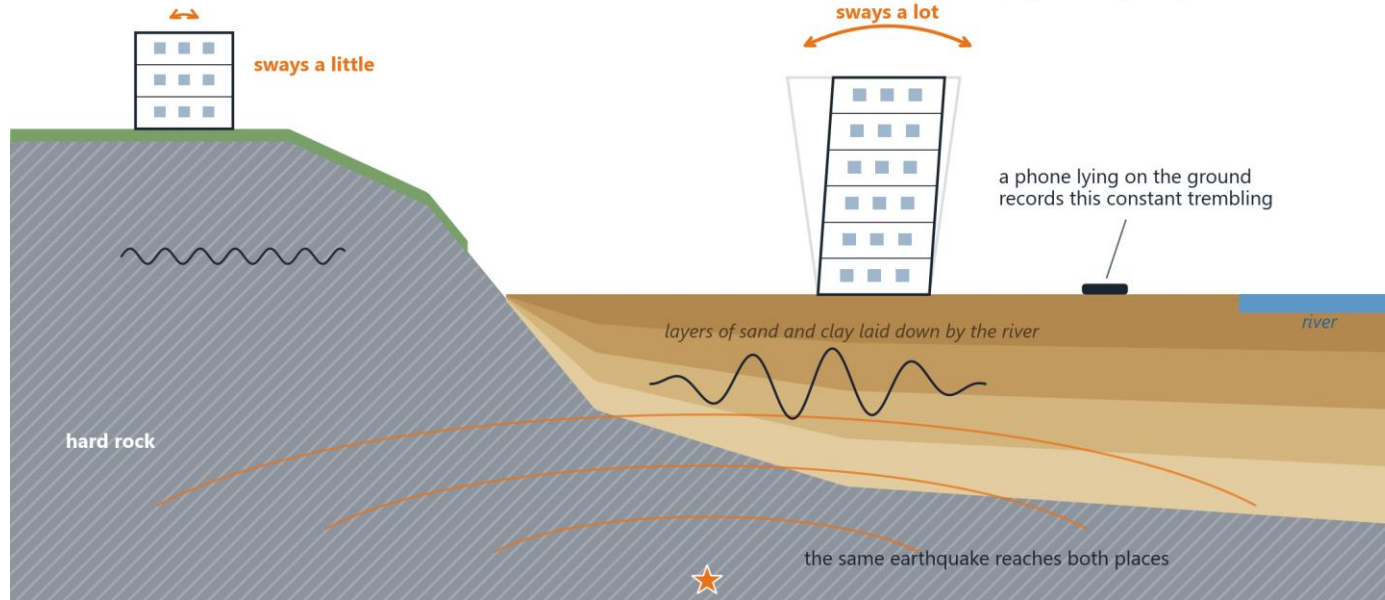


Real vibration our phone recorded at the five spots, one minute each, 26 September 2026

Two buildings can feel the same earthquake very differently, depending on the ground beneath them. Which parts of our campus sit on the riskier ground?

FIRM GROUND (HILLS)

Stiff rock passes the shaking on without letting it build up.



SOFT, DEEP RIVER SOIL

Shaking gets trapped between the soil and the rock below and builds up, like jelly wobbling on a plate.

WHY IT MATTERS HERE

Zone V

Assam is in India's highest earthquake-risk zone. In 1897 the shaking turned parts of Guwahati's riverside soil to slush.

1985

In Mexico City, soft lake-bed soil swayed in step with 10 to 14-storey buildings, and those were the ones that fell.

0.2–10

sways per second: how much Guwahati's ground rhythm changes from floodplain to hill. The city map is too coarse to tell campus blocks apart.

WHO NEEDS THE ANSWER

Estate Office

Decides where new buildings go, and pays ₹5,000 to ₹15,000 for every metre of soil drilling.

Civil engineers

Study earthquakes, but have no side-by-side ground measurements for the campus.

HOW A PHONE MIGHT TELL THE DIFFERENCE

Soft soil sways sideways more than it bounces, at a rhythm of its own, like jelly on a plate. Comparing the two directions reveals that rhythm. Geologists call it the H/V method (Nakamura, 1989).

1

Traffic, wind and footsteps keep the ground trembling a tiny bit, all the time.

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2

A phone lying flat records the sideways and the up-and-down shaking.

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3

Where sideways ÷ up-and-down peaks is the ground's rhythm: slower means softer, deeper soil.

OUR MOTIVATION

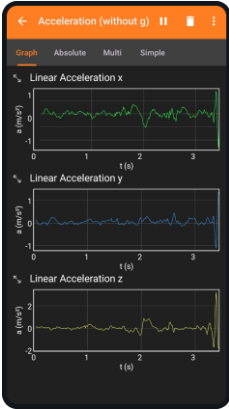
Drilling tells you the truth about the ground, but it is expensive and slow. We wanted to see whether a few quick vibration recordings, made with the simplest tools, could screen the whole campus first and point out which spots sit on softer ground that is more likely to be damaged in an earthquake.

We turned one phone into a vibration recorder, took it to five spots on campus in one evening, and built checks that refuse to trust weak data.



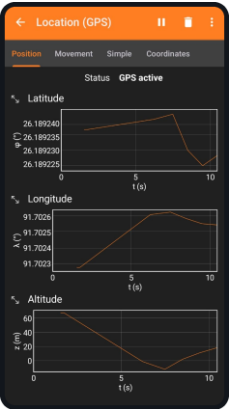
THE TOOL: JUST A PHONE

We deliberately picked the simplest measuring tool there is, rather than a lab instrument, to see how far everyday hardware can go. The free phyphox app (RWTH Aachen University) reads the phone's sensors directly, and at every spot we ran three of its tools in the same order.



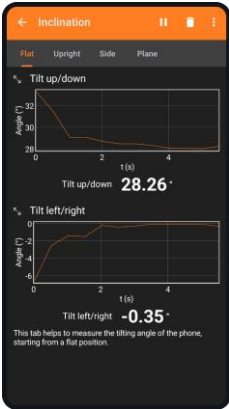
1 · VIBRATION

The phone lies flat on bare ground, untouched, for about three and a half minutes, recording shaking in three directions about 100 times a second. The big spikes at the right edge are just the phone being picked up.



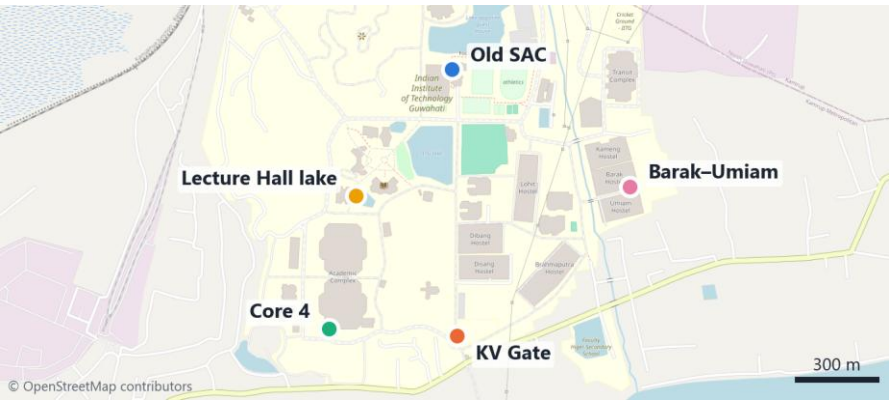
2 · LOCATION

A GPS reading ties each recording to a place. Four of the five spots got a precise satellite fix, accurate to about 2 m. At Barak-Umiam the phone only managed a rough fix (± 100 m), and we say so wherever that spot appears.



3 · TILT CHECK

If the phone isn't lying flat, sideways and up-and-down shaking get mixed together. Four spots were within 5° of level; KV Gate was 5.6° , so its result carries a warning. The 28° here is the phone being held up for this picture.



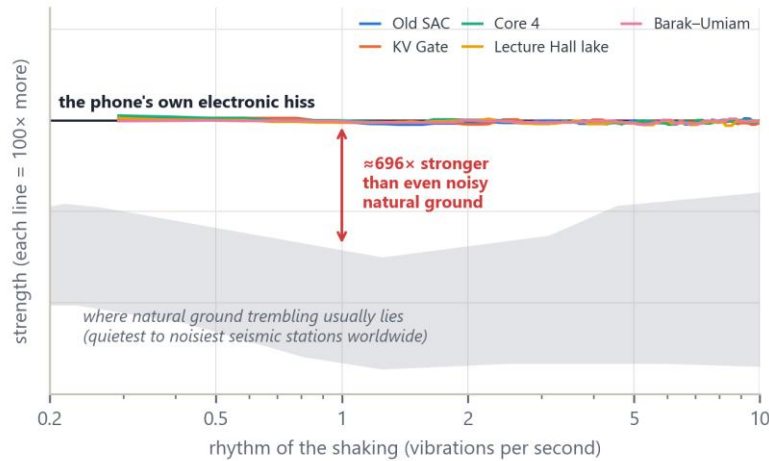
FROM 100,000 READINGS TO AN ANSWER, IN FOUR STEPS

- 1 Remove the moments we disturbed.** We cut the first and last 5 seconds, when a hand was near the phone, and any stretch where footsteps or a passer-by made the readings jump. 80 of the 93 twenty-second pieces were clean enough to keep.
- 2 Find each spot's rhythm.** For every clean piece we compared sideways and up-and-down shaking at each rhythm from 0.5 to 15 vibrations per second, smoothed the result in the standard way, and averaged the pieces together.
- 3 Test it against the rulebook.** A European research project (SESAME, 2004) lists nine checks a real ground rhythm has to pass, such as a clear peak that shows up in the same place in every piece. We ran all nine.
- 4 Make sure the phone can hear the ground.** Every sensor has a faint electronic hiss of its own. If the hiss is louder than the ground, the ratio only describes the phone. We added this check after it caught a mistake in our own first result.

WHERE AND WHEN

We picked five spots on different kinds of ground: beside the lake near the lecture halls, the academic complex near Core 4, the Old SAC area, the KV school gate, and the open ground between Barak and Umiam hostels. All five were recorded on 26 September 2026 between 7:57 and 10:49 pm. We chose night on purpose: the ground is quietest then, which makes it the hardest test for a phone. That gave 15 recordings, about 17 minutes of vibration and just over 100,000 individual readings.

In 3½ quiet night-time minutes the phone couldn't hear the ground's slow trembling, but it caught real vibration, and longer recordings can go further.



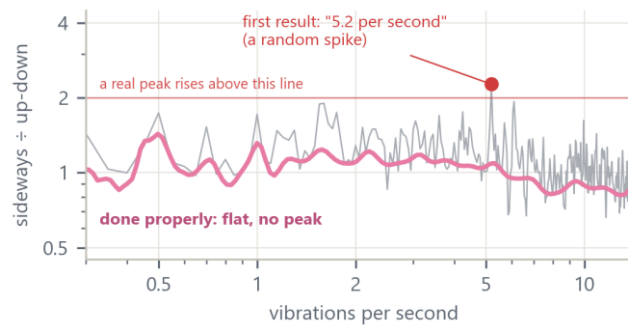
38–696× **80 $\mu\text{m/s}$**

how much stronger the phone's own hiss is than natural ground trembling

the most any spot vibrated at night, under the operating-theatre limit

Every coloured line (one per spot) sits on the phone's flat hiss, far above the grey band where natural trembling lives. Like hearing a whisper beside a running fan.

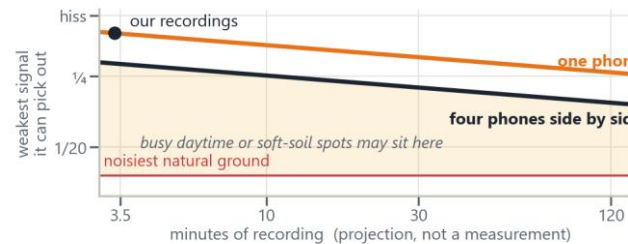
- ✓ Captured footsteps, passers-by and a pump switching on
- ✓ Measured the phone's own noise level, the same at every spot
- ✗ Slow ground rhythms need longer, multi-phone recordings



THE MISTAKE WE CAUGHT, AND WHAT AI SAID

Our first attempt found a "rhythm" of 5.2 per second at Barak-Umiam (grey), a random spike in unsmoothed data; done properly (pink), the curve is flat. An AI model trained to tell the spots apart by their vibration recognised the spot with the pump, but found no difference in the ground elsewhere (21% right, against 20% by chance).

HOW FAR LONGER RECORDINGS CAN GO



WHAT THIS MEANS TODAY

Civil engineers: a tested routine, five mapped spots and the phone's measured noise level, so the next round can be planned properly.

Estate Office: all five spots are quieter at night than hospital operating-theatre limits, and a pump was flagged near Barak-Umiam.

WHAT COMES NEXT

- 1 **Longer recordings.** 30 minutes in daylight with two phones side by side at the riverside spot: the cheapest way to see how far phones can go.
- 2 **Better sensor.** A ₹50,000 to ₹1.2 lakh geophone at dozens of spots, same routine.
- 3 **Train AI.** On those readings plus the boreholes campus already has, to predict ground quality in between, map the whole campus with its confidence, and show where to drill next.

WHERE IT CAN GO

Campuses

Builders & townships

Insurers

A cheap ground screen first; ₹1.5 to 4.5 lakh boreholes only where they matter.

What's real and what's calculated. Every number comes from real phone recordings: 100,212 vibration readings, 94 GPS fixes and 210 tilt readings. The ground rhythms, rulebook checks and AI results are calculated from those readings, the longer-recording chart is a projection from them, and nothing is simulated.

GitHub Link (Data and Codes): github.com/agayushhh/Granica_x_Sismic_solution_AI **Drive Link:** drive.google.com